



Chapter 1 Number System 1

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NUMBER SYSTEM

Learning Objectives

1. Classifying the sets of real number.
2. Manipulate the expression of index and surd.
3. Solve equation involving index, surd and logarithm.
4. Writing real numbers in the form of set, interval, inequality and line number.

Keywords : Real number, rational number, irrational number, whole number, natural number, integers, interval, line number, index, surd, logarithm

1.1 Real Number

The concept of number is the most important fundamental in mathematics and consequently in science. But the concept of number was not acceptably complete until the late 1800s when mathematicians started to use the concept of “set” to explain numbers. In this chapter, we will learn about the set of real number.

In mathematics, real number is the union or combination subsets that is the rational numbers and irrational numbers. The real number is usually denoted as $\mathbb{R}$.

The rational number is denoted as $\mathbb{Q}$, while the irrational number is denoted as $\mathbb{H}$. Another symbol is also used to write irrational number in term of complement set of $\mathbb{Q}$, that is $\mathbb{Q}^c$.

Moreover, in the set of rational number, there are many subsets of numbers.

- (i) Natural number, $\mathbb{N}$

The set of natural numbers denoted $\mathbb{N}$ is defined as

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

- (ii) Whole number, W

Natural number together with 0 form the set of whole number, W .

$$W = \{0, 1, 2, 3, 4, \dots\}$$

- (iii) Integer, $\mathbb{Z}$

The set of integers, $\mathbb{Z}$ consists of natural numbers, zero and the negative of natural numbers.

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

The set of positive integers is

$$\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\},$$

The set of negative integers is

$$\mathbb{Z}^- = \{\dots, -3, -2, -1\}.$$

The positive integers $2, 4, 6, 8, \dots$ are called even numbers and can be represented as $\{2n, n \in \mathbb{N}\}$. The negative integers $1, 3, 5, 7, \dots$ are called odd numbers and can be represented generally as $\{2n+1, n \in \mathbb{N}\}$. A prime number is a natural number greater than 1 which has 1 and itself as factors. For example $2, 3, 5, 7, 9, 11, 13, 15, 17, 19, \dots$ are prime numbers.

(iv) Rational number, $\mathbb{Q}$

Number that can be written in the form $\frac{p}{q}$, also known as fractions, where $p, q \in \mathbb{Z}$ and $q \neq 0$ are called rational numbers, $\mathbb{Q}$

$$\mathbb{Q} = \{x : x = \frac{p}{q}, p, q \in \mathbb{Z}, q \neq 0\}.$$

The decimal representations of rational numbers are repeating or terminating. For example,

$$\left. \begin{array}{l} \frac{3}{5} = 0.6 \\ \frac{7}{4} = 1.75 \end{array} \right\} \text{Terminating}$$

$$\frac{1}{6} = 0.1666\dots \text{“6” is repeating and can be written as } 0.1\bar{6}$$

$$\frac{4}{3} = 1.3333\dots \text{“3” is repeating and can be written as } 1.\bar{3}$$

$$\frac{3}{11} = 0.272727\dots \text{“27” is repeating and can be written as } 0.2\bar{7}$$

(v) Irrational number, $\mathbb{Q}'$

Number that cannot be written as fractions and whose decimals neither terminate nor repeat are called irrational numbers and are denoted by $\mathbb{Q}'$ or $\bar{\mathbb{Q}}$. For example,

$$\sqrt{2} = 1.41421356\dots$$

$$\pi = 3.14159265\dots$$

$$e = 2.71828183\dots$$

All rational and irrational numbers constitute the set of real number, $\mathbb{R}$.

From Figure 1.1, we can conclude that

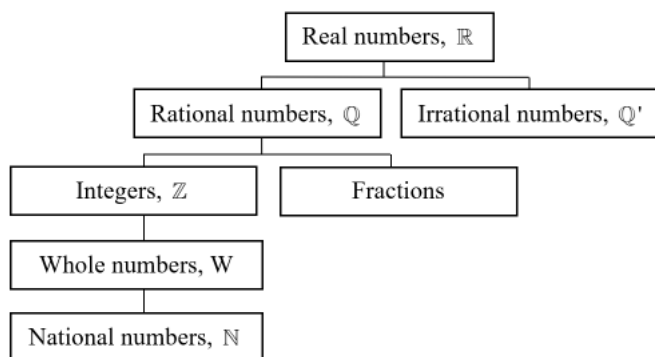


Figure 1.1: The real number system

- (i) $\mathbb{Q} \cup \mathbb{Q}' = \mathbb{R}$
- (ii) $\mathbb{Q} \cap \mathbb{Q}' = \emptyset$
- (iii) $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$

Example 1.1 Write the following as fraction

(a) $1.\bar{2}$

Solution

Let	$x = 1.2222\dots$
Multiply both sides by 10	$10x = 12.2222\dots$
Subtract	$9x = 11$
	$x = \frac{11}{9}$

(b) $0.\overline{35}$

Solution

Let	$x = 0.353535\dots$
Multiply both sides by 100	$100x = 35.353535\dots$
Subtract	$99x = 35$
	$x = \frac{35}{99}$

Practice 1.1 Express each of the following rational numbers as a fraction.

(a) $0.6\bar{3}$

(c) $1.\bar{3}$

(b) $0.41\bar{6}$

(d) $0.8\bar{3}$

Answer :

(a) $\frac{7}{11}$

(b) $\frac{5}{12}$

(c) $\frac{4}{3}$

(d) $\frac{5}{6}$

Example 1.2 Based on the list of real numbers given below

$$18, \sqrt{10}, 11, 9, 4.0, \frac{1}{3}, 0, 6, -1, 44, \sqrt{8}, 0.9999 \dots, \sqrt{2}, \frac{1}{7}, 1.2345 \dots,$$

identify the set of

- (a) natural numbers,
- (b) whole numbers,
- (c) integers,
- (d) rational numbers,
- (e) and, irrational numbers.

Solution

(a) The set of natural numbers = $\{18, 11, 9, 6, 44\}$.

(b) The set of whole numbers = $\{18, 11, 9, 0, 6, 44\}$.

(c) The set of integers = $\{18, 11, 9, 0, 6, -1, 44\}$.

(d) The set of rational numbers

$$= \{18, 11, 9, 4.0, \frac{1}{3}, 0, 6, -1, 44, 0.9999 \dots, \frac{1}{7}\}.$$

(e) The set of irrational numbers = $\{\sqrt{10}, \sqrt{8}, \sqrt{2}, 1.2345 \dots\}$.

Practice 1.2 Classify the number $\sqrt{2}$ as natural numbers, $\mathbb{Q}$ or irrational number $\mathbb{Q}'$.

- | | | |
|--------------------|--------------------|------------------|
| (a) $\frac{-2}{5}$ | (d) e^2 | (g) 3.142 |
| (b) 5π | (e) $\frac{22}{7}$ | (h) 0 |
| (c) $5.1\bar{44}$ | (f) $\sqrt{2}$ | (i) 5.3215467... |
| | | (j) 0.010101... |

Practice 1.3 State whether the following statements are true or false.

- (a) $\frac{e}{2}$ is a rational.
- (b) $\sqrt{-4}$ is a real number.
- (c) $2\sqrt{2}$ is an irrational number.
- (d) 1.12345 is an irrational number.
- (e) Every irrational number is a real number.
- (f) $\mathbb{Q} \cap \mathbb{Q}' = \emptyset$.
- (g) Every rational number is an integer.
- (h) Every prime number is a natural number.
- (i) $\sqrt{2} - 1$ is a rational number.
- (j) $\mathbb{Z} \cap \mathbb{N} = W$.

Practice 1.4 Based on the list of real numbers given below,

$$13, \sqrt{10}, 11, 9, 4.0, \frac{1}{3}, \sqrt{8}, 0.9999\dots, \frac{1}{7}, 1.2345\dots$$

Identify the set of

- (a) natural numbers,
- (b) whole numbers,
- (c) integers,
- (d) rational numbers,
- (e) irrational numbers.

1.2 Real Number Line

Real numbers can be represented on a straight line called the real number line, Every point on the real number line corresponds uniquely to a real number,

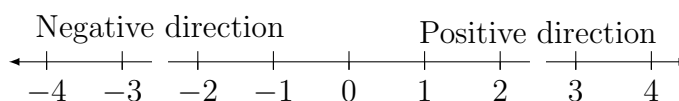


Figure 1.2: Real number line

A point called origin represents zero. Point to the right of the origin represent positive numbers while points to the left of the origin represent negative numbers.

1.3 Intervals of Real Numbers

Finite intervals

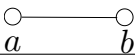
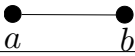
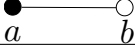
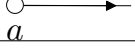
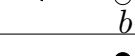
If a and b are real numbers such that $a < b$, then

- (i) $\{x : a < x < b\}$ is called the open interval from a to b .
- (ii) $\{x : a \leq x \leq b\}$ is called the closed interval from a to b .
- (iii) $\{x : a \leq x < b\}$ or $\{x : a < x \leq b\}$ is called the half-open or half-closed interval from a to b .

Infinite intervals

If a is a real number, then the set of all real numbers x satisfying any of the conditions $x < a$, $x > a$, $x \leq a$ or $x \geq a$ is called an infinite interval.

Table 1.1 Sets of real number in interval notation and its representation in real number line.

Interval	Set notation	Representation on a real number line
(a, b)	$\{x : a < x < b\}$	
$[a, b]$	$\{x : a \leq x \leq b\}$	
$(a, b]$	$\{x : a < x \leq b\}$	
$[a, b)$	$\{x : a \leq x < b\}$	
(a, ∞)	$\{x : x > a\}$	
$[a, \infty)$	$\{x : x \geq a\}$	
$(-\infty, b)$	$\{x : x < b\}$	
$(-\infty, b]$	$\{x : x \leq b\}$	
$(-\infty, \infty)$	$\{x : x \in \mathbb{R}\}$	

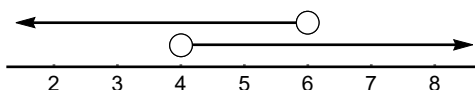
Example 1.3

Given the following sets. Find the interval notation and draw them on a number line.

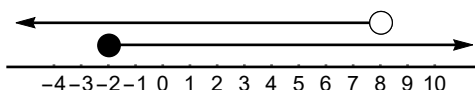
- (a) $\{x : 4 < x < 6\}$.
- (b) $\{x : -2 \leq x < 8\}$.
- (c) $\{x : 7.5 < x \leq 11\}$.
- (d) $\{x : -9 \leq x \leq -3\}$.

Solution

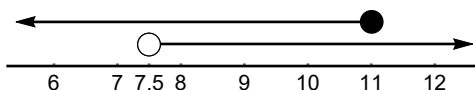
(a) $\{x : 4 < x < 6\} = (4, 6)$



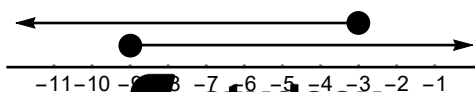
(b) $\{x : -2 \leq x < 8\} = [-2, 8)$



(c) $\{x : 7.5 < x \leq 11\} = (7.5, 11]$



(d) $\{x : -9 \leq x \leq -3\} = [-3, 9]$



Practice 1.5

Complete the following table.

(a)	No.	Representation	Interval	Inequality
	(i)		$[-1, 5]$	
	(ii)		$[3, 7]$	
	(iii)			$x \leq -4$
	(iv)		$(0, \infty)$	
	(v)		$(-3, 8)$	

(b) By using number lines, simplify each of the following.

(i) $(-5, 8) \cup [2, \infty)$

(iii) $[-6, 1) \cap (-2, 7)$

(ii) $(-8, -1) \cup (-2, 6]$

(iv) $(-\infty, 8) \cap (3, 11]$

(c) Given that $(-2, 2) \cup (-1, m] = (n, 6]$. Find the values of m and n .

(d) If $P = (-8, 1]$, $Q = [-6, 3]$ and $R = (-9, -1)$, find

(i) $P \cap Q \cap R$

(ii) $P \cup Q \cup R$

1.4 Indices

If a is a real number and n is a positive integer, then

$$a^n = \underbrace{a \times a \times \dots \times a}_{n \text{ times}}$$

The number a is called the base and n is called the index (also known as exponent). a^n is read as ' a to the power n '.

Laws of indices

If $m, n \in \mathbb{Q}$ and $a, b \in \mathbb{R}$, $a \neq 0$, $b \neq 0$ then

- (i) $a^m \times a^n = a^{m+n}$ (product rule)
- (ii) $a^m \div a^n = a^{m-n}$ (quotient rule)
- (iii) $(a^m)^n = a^{mn}$ (power of a power rule)
- (iv) $a^0 = 1$
- (v) $a^{-n} = \frac{1}{a^n}$ (definition of negative index)
- (vi) $(ab)^n = a^n b^n$ (power of product rule)
- (vii) $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ (power of quotient rule)

Example 1.4

$$2^4 = 2 \times 2 \times 2 \times 2 = 16$$

$$3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243$$

$$\left(\frac{1}{2}\right)^2 = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$(-6)^2 = -6 \times -6 = 36$$

$$-5^4 = -(5 \times 5 \times 5 \times 5) = -625$$

Practice 1.6 Without using calculator, evaluate

- (a) 2^5
- (c) $27^{-\frac{1}{3}}$
- (e) $0.008^{\frac{2}{3}}$
- (b) $81^{\frac{1}{4}}$
- (d) $0.01^{-\frac{1}{2}}$
- (f) $\left(-\frac{1}{5}\right)^{-3}$

Example 1.5 The following are the examples of the usage of law of indices

(a) $2^2 \cdot 2^3 = 2^{2+3} = 2^5 = 32.$

(b) $(4^3)^2 = 4^6 = 4096.$

(c) $\pi^{-4} \cdot \pi^{-6} = \pi^{-4-6} = \pi^{-10}.$

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$$(d) \quad (-2a)^{-3} = \frac{1}{(-2a)^3} = \frac{-1}{8a^3}.$$

$$(e) \quad \left(\frac{m^4}{n^5}\right)^8 = \frac{m^{32}}{n^{40}}.$$

$$(f) \quad (3y^3)(5y^5) = 15y^{3+5} = 15y^8.$$

$$(g) \quad \frac{1}{(m-n)^{-4}} = 1 \div \frac{1}{(m-n)^4} = (m-n)^4.$$

$$(h) \quad \left(\frac{r}{3s^{-8}}\right)^4 = \frac{r^4}{(3s^{-8})^4} = \frac{r^4}{81s^{-32}}.$$

$$(i) \quad \left(\frac{x^4(y-1)}{(x^2+y^2)}\right)^4 = \frac{x^{16}(y-1)^4}{(x^2+y^2)^4}.$$

Practice 1.7 Simplify

$$(a) \quad \frac{x^{-2}x^{\frac{2}{3}}}{x^{-\frac{1}{3}}}$$

$$\text{Ans : } (c) \quad \frac{9x^2y^5}{6y^{-3}} \left(\frac{1}{xy^{-4}}\right)^2$$

$$(b) \quad \frac{(2u^3v^{-4})^2}{8uv^{-6}}$$

$$(d) \quad \frac{(y-1)^{\frac{1}{2}} + (y-1)^{-\frac{1}{2}}}{(y-1)^{\frac{1}{2}}}$$

Answer :

$$(a) \quad \frac{1}{x^{5/3}}$$

$$(b) \quad \frac{32u^7}{v^{14}}$$

$$(c) \quad 54y^{10}$$

$$(d) \quad \frac{y}{y-1}$$

Example 1.6

(a) Solve the equation $64^x = 4$.

Solution

Given

$$64^x = 4,$$

Can be written as

$$(4^3)^x = 4,$$

From the law of index

$$4^{3x} = 4,$$

equating the exponent

$$3x = 1,$$

and finally we obtained

$$x = \frac{1}{3}.$$

(b) Solve the equation $9^x - 6(3)^x + 5 = 0$

Solution

Given

Notice that

Therefore it can be written as

By letting $y = 3^x$,

Solving the quadratic equation to get

Next we substitute back

to get

$$9^x - 6(3)^x + 9 = 0,$$

$$9^x = (3^2)^x = 3^{2x} = (3^x)^2,$$

$$(3^x)^2 - 6(3^x) + 9 = 0,$$

$$y^2 - 6y + 9 = 0,$$

$$(y - 3)(y - 3) = 0,$$

$$y = 3,$$

$$3^x = 3 \text{ or } 3^x = 3^1,$$

$$x = 1.$$

Practice 1.8 Solve the following equations:

(a) $5^{x-2} = 625$

(e) $3^{2x+2} + 3^{x+3} - 10 = 0$

(b) $3^{3y-1}9^{y-1} = 81^{2y}$

(f) $2^{2x+1} + 4 = 9(2^x)$

(c) $4^{(2x-1)^2} = 2^{x+1}$

(g) $2^{2t-1} + 2^{t+1} = 16$

(d) $100^{x^2} = 10^{5x-3}$

(h) $9(3^{2x}) - 28(3^x) + 3 = 0$

Answer :

(a) 6

(d) $1, \frac{3}{2}$

(g) 2

(b) -1

(e) -1

(c) $\frac{1}{8}, 1$

(f) -1, 2

(h) -2, 1

1.5 Surds

If b is a positive integer, i.e $b > 0$ then there is only one positive real number t such that $t^n = b$. This positive number is the principle n^{th} root. Since

we have

$$\underbrace{b^{\frac{1}{n}} \times b^{\frac{1}{n}} \times b^{\frac{1}{n}} \times \dots \times b^{\frac{1}{n}}}_{n \text{ times}} = b.$$

Therefore, $b^{\frac{1}{n}}$ is the n^{th} root of b , and this $b^{\frac{1}{n}}$ is the principal n^{th} root. This principal n^{th} root of b can be written in another form as $\sqrt[n]{b}$ (i.e. $b^{\frac{1}{n}} = \sqrt[n]{b}$). $b^{\frac{1}{n}}$ is in index or exponential form and $\sqrt[n]{b}$ is in radical form. The symbol $\sqrt{}$ is called a radical, b is the radicant and n is the index.

If $n = 2$, we write $b^{\frac{1}{2}} = \sqrt{b}$. In general, $b^{\frac{m}{n}} = (b^m)^{\frac{1}{n}} = \sqrt[n]{b^m}$.

Remarks

The number $\sqrt{b} = b^{\frac{1}{2}}$, $b > 0$ represent only one number.

For example,

$$\sqrt{9} = 9^{\frac{1}{2}} = 3$$

and

$$-\sqrt{9} = -(9^{\frac{1}{2}}) = -3.$$

It is wrong to write $\sqrt{9} = \pm 3$.

A surd is an expression containing a root with an irrational solution. For example, $\sqrt{5}$, $2 - \sqrt{3}$ and $\sqrt{7} + \sqrt{11}$. When a fraction involves a surd in the denominator, we usually rationalize the fraction, which is we get rid of the radical sign in the denominator.

Usually this can be done by multiplying the denominator (and numerator) by the conjugate of the denominator.

For example,

$$\begin{aligned} \frac{1}{3 + \sqrt{2}} &= \frac{1}{3 + \sqrt{2}} \times \frac{3 - \sqrt{2}}{3 - \sqrt{2}} \\ &= \frac{3 - \sqrt{2}}{9 - 2} = \frac{3 - \sqrt{2}}{7} \end{aligned}$$

Notice that the denominator of $\frac{3 - \sqrt{2}}{7}$ does not contain surds anymore.

Practice 1.9 Rationalize the denominator for each of the following surds.

(a) $\frac{9}{\sqrt{13}}$

(d) $\frac{1}{\sqrt{5} - \sqrt{5}}$

(b) $\frac{1}{\sqrt[5]{2}}$

(e) $\frac{3 + \sqrt{2}}{5 - \sqrt{2}}$

(c) $\frac{2\sqrt{3}}{\sqrt{5} - 2}$

(f) $\frac{9 + 5\sqrt{2}}{3 + 2\sqrt{2}}$

(a) $3\sqrt{3}$

(c) $2\sqrt{15} + 4\sqrt{3}$

(e) $\frac{17 + 8\sqrt{2}}{23}$

(b) $\frac{\sqrt[5]{16}}{2}$

(d) $\frac{\sqrt{7} + \sqrt{5}}{2}$

(f) $7 - 3\sqrt{2}$

Practice 1.10 Without using calculator, simplify:

(a) $\sqrt{\frac{3}{5}} \sqrt{\frac{75}{27}}$

(c) $\frac{3\sqrt{12}}{4\sqrt[3]{54}}$

(d) $(2\sqrt{3} - \sqrt{2})(1 - 2\sqrt{6})$

(b) $8\sqrt{2} - \sqrt{32} + 5\sqrt{98}$

(e) $(\sqrt{5} - \sqrt{2})^2$

Answer:

(a) $\frac{\sqrt{15}}{3}$

(c) $\frac{\sqrt{3}\sqrt[3]{4}}{4}$

(e) $7 - 2\sqrt{10}$

(b) $39\sqrt{2}$

(d) $-2\sqrt{3} - 13\sqrt{2}$

Practice 1.11 Simplify the following expressions:

(a) $\sqrt{\frac{x}{y}} \sqrt{\frac{25y}{x^3}}$

(c) $\sqrt{\sqrt[3]{64x^8y^{27}}}$

(b) $\sqrt{x^8y^2z^9}$

(d) $\frac{2 - \sqrt{3}}{2 + \sqrt{3}} - \frac{2 + \sqrt{3}}{2 - \sqrt{3}}$

Answer:

(a) $\frac{5}{x}$

(b) $x^4yz^4\sqrt{z}$

(c) $\sqrt{2xy}\sqrt[4]{x}\sqrt{y}$

(d) $-8\sqrt{3}$

Example 1.7

(a) Solve $\sqrt{4x+5} = 5$.

Solution

$$\begin{array}{rcl}
 \sqrt{4x+5} & = & 5 \\
 \text{Squaring both sides,} & 4x+5 & = 25 \\
 & 4x & = 20 \\
 & x & = 5.
 \end{array}$$

(b) Solve $\sqrt{x} + 2 = \sqrt{x+8}$.

Solution

$$\begin{array}{rcl}
 \sqrt{x} + 2 & = & \sqrt{x+8} \\
 \text{Squaring both sides,} & \sqrt{x} & = \sqrt{x+8} - 2 \\
 & x & = (x+8) - 4\sqrt{x+8} + 4 \\
 & -12 & = -4\sqrt{x+8} \\
 & x+8 & = 9 \\
 & x & = 1.
 \end{array}$$

(c) $\sqrt{x+2} + \sqrt{x-5} = \sqrt{3x+7}$.

Solution

$$\begin{aligned}
\sqrt{x+2} + \sqrt{x-5} &= \sqrt{3x+7} \\
(x+2) + 2\sqrt{x+2}\sqrt{x-5} + (x-5) &= 3x+7 \\
2\sqrt{x+2}\sqrt{x-5} &= x+10 \\
4(x+2)(x-5) &= (x+10)^2 \\
3x^2 - 32x - 140 &= 0 \\
(x-14)(3x+10) &= 0 \\
x = 14 \text{ or } x &= -\frac{10}{3}.
\end{aligned}$$

We need to check if both solution fits with the original problem since the solution obtained is actually the solution for the quadratic equation. Hence we need to check with the original question by replacing $x = 14$ and $x = -\frac{10}{3}$ into $\sqrt{x+2} + \sqrt{x-5} = \sqrt{3x+7}$. Finally we found that only $x = 14$ can satisfy the original problem. Therefore $x = 14$ is the only solution.

Practice 1.12 Solve the following equations.

- | | |
|---------------------------------|---|
| (a) $\sqrt{3x-2} = 4$ | (d) $\sqrt{x} - \sqrt{x+2} = 1$ |
| (b) $x+5 = \sqrt{7-x}$ | (e) $\sqrt{x+7} = \sqrt{x+2} + \sqrt{2x-3}$ |
| (c) $\sqrt{x} + 1 = \sqrt{x+7}$ | (f) $\sqrt{3x+1} + 2 = \sqrt{x-1}$ |

Answer:

- | | | |
|-------------|-----------------|-----------------|
| (a) $x = 6$ | (c) $x = 9$ | (e) $x = 2$ |
| (b) $x = 2$ | (d) no solution | (f) no solution |

1.6 Logarithms

In exponential form, $2^3 = 8$, where 2 is the base and 3 is the exponent. In logarithmic form, this can be expressed as $\log_2 8 = 3$. This is read

as ‘logarithmic 8 to base 2 is 3’.

Similarly,

$$3^{-2} = \frac{1}{9} \text{ or } \log_3 \left(\frac{1}{9} \right) = -2.$$

In general, if a and y are positive real numbers such that $a^x = y$, the logarithm of y to base a is x .

$$a^x = y \text{ if and only if } \log_a y = x \text{ where } a, y \in \mathbb{R}^+.$$

Log with base 10 is usually written as $\log x$ where the base 10 is omitted. In science and engineering, Log with base e is usually used where $e = 2.718281828450 \dots$. It is written as $\ln x$ where x is any positive number.

Practice 1.13 Without using table or calculator, evaluate:

- | | |
|---------------------------|-------------------------|
| (a) $\log_3 81$ | (e) $\log_2 32$ |
| (b) $\log_4 \frac{1}{16}$ | (f) $\log_3 27\sqrt{3}$ |
| (c) $\log_{10}(0.001)$ | (g) $\log_{0.5} 0.0625$ |
| (d) $\log_8 128$ | (h) $\log_4 0.25$ |

Answer:

- | | | | |
|--------|-------------------|-------------------|--------|
| (a) 4 | (c) -3 | (e) 5 | (g) 4 |
| (b) -2 | (d) $\frac{7}{3}$ | (f) $\frac{7}{2}$ | (h) -1 |

Laws of logarithm

If a, x, y are positive real numbers, then

- (i) $\log_a xy = \log_a x + \log_a y$
- (ii) $\log_a \frac{x}{y} = \log_a x - \log_a y$
- (iii) $\log_a 1 = 0$

$$(iv) \log_a a = 1$$

$$(v) \log_a x^c = C \log_a x, C \text{ is any real number.}$$

$$(vi) a^{\log_a x} = x.$$

Practice 1.14 Express each of the following as a single logarithm.

$$(a) \log_a x + \frac{1}{2} \log_a y$$

$$(b) 2 \log_2 5 + \frac{1}{2} \log_2 9 - \log_2 \left(\frac{1}{4} \right)$$

$$(c) 2 \log_a x + \log_a y - \frac{1}{2}$$

$$(d) \log_x 3 + \log_x 35 + \log_x \left(\frac{15}{35} \right) - \log_x \left(\frac{7}{24} \right)$$

$$(e) \frac{1}{3} \log_a x - 9 \log_a y - 6 \log_z \mathbb{Z}$$

$$(f) \frac{1}{4} (3 \log_a x + 5 \log_a y)$$

Answer:

$$(a) \log_a xy^{\frac{1}{2}}$$

$$(c) \log_a \frac{x^2 y}{\sqrt{a}}$$

$$(e) \log_a \frac{x^{\frac{1}{3}}}{y^9 z^6}$$

$$(b) \log_2 300$$

$$(d) \log_x 270$$

$$(f) \log_a x^{\frac{3}{4}} y^{\frac{5}{4}}$$

Changing the base of logarithms

The formula to change the base of a logarithm from a to b can be derived as follows:

If $\log_a c = u$ then $c = a^u$

$$\begin{aligned}\log_b c &= \log_b a^u \\ &= u \log_b a\end{aligned}$$

$$u = \frac{\log_b c}{\log_b a}$$

$$\text{Therefore, } \log_a c = \frac{\log_b c}{\log_b a}$$

Remarks

When $c = b$

$$\begin{aligned}\log_a b &= \frac{\log_b b}{\log_b a} \\ &= \frac{1}{\log_b a}.\end{aligned}$$

Practice 1.15 By changing the base to 10, evaluate the following logarithm. Write your answer to 4 decimal points.

(a) $\log_5 15$

(c) $\log_2 75$

(b) $\log_{0.5} 7$

(d) $\log_8 1.25$

Answer:

(a) 1.6826

(c) 6.2288

(b) -2.8074

(d) 0.1073

Logarithmic Equations

Equation that contains logarithm of an unknown are called logarithmic equations. For examples,

$$\log_2 (3x - 1) = 3, \quad \log_4 x - 6 + \log_4 x = 2.$$

The properties of logarithms are used to simplify and solve logarithmic equations. Each solution obtained must be checked and values which do not satisfy the given equation are rejected. Logarithms are not defined for negative numbers.

Example 1.8

(a) Solve $\log_2(x + 1) + \log_2(x - 2) = 2$.

Solution

$$\log_2(x + 1) + \log_2(x - 2) = 2$$

$$\log_2(x + 1)(x - 2) = 2$$

$$(x + 1)(x - 2) = 4$$

$$x^2 - x - 6 = 0$$

$$(x + 2)(x - 3) = 0$$

$$x = -2 \text{ or } x = 3.$$

Replacing $x = -2$ and $x = 3$ into the original problem, we found that only $x = 3$ is suitable. Hence the solution is $x = 3$.

(b) If $10^x + 10^{-x} = 4$, prove that $x = \log_{10}(2 \pm 3)$.

Solution

Let $10^x = p$. Then $\frac{1}{p} = 10^{-x}$

Hence, $10^x + 10^{-x} = p + \frac{1}{p} = 4$

$$p^2 + 1 = 4p$$

$$p^2 - 4p + 1 = 0$$

Using quadratic formula, $p = \frac{4 \pm \sqrt{16 - 4}}{2}$

$$p = 2 \pm \sqrt{3}.$$

Therefore, $10^x = 2 \pm \sqrt{3}$

$$x = \log_{10}(2 \pm \sqrt{3}).$$

Practice 1.16 Solve the following logarithmic and index equations.

(a) $\log_6(x + 3) + \log_6(x - 2) = 1.$

(b) $\log_2(5x - 7) = 3.$

(c) $\log(2t^3 + 1) = 3 \log t.$

(d) $\ln(x + 1) + \ln x = \ln(3x - 1).$

(e) $5^{2x} = 3(2^x).$

(f) $4^x - 2^{x+1} - 3 = 0.$

Answer:

(a) $x = 3$

(c) $t = \frac{1}{2}$

(e) $x = 0.43497$

(b) $x = 3$

(d) $x = 1$

(f) $x = 1.58496$

Practice 1.17 Solve the following logarithmic equations.

(a) $\log_5 x = 2$

$$(b) \ln(x - 5) = 1$$

$$(c) \log x = x \log 3 + \log(5x - 2)$$

$$(d) \log_3(3t) - \frac{1}{2} \log_3 4 = \lg_3(t - 1)$$

$$(e) (\log x)^2 = \log x^3$$

$$(f) 5 - 4 \log_3 x = \log_x 3$$

$$(g) 2x - 2^{\log x} = 0$$

Answer:

$$(a) 25$$

$$(d) \text{ no solution}$$

$$(b) 5 + e$$

$$(e) 4$$

$$(f) 1, 10^2$$

$$(c) \frac{9}{22}$$

$$(g) 3, 1.315$$

1.7 Exercise 1

1. State whether each of the following statement is true or false.

- (a) All whole numbers are integers.
- (b) All integers are natural numbers.
- (c) $\sqrt{7}$ is a rational number.
- (d) 4.58 is a rational number.
- (e) $0.121212\dots$ is a irrational number.

2. Set A is given as

$$A = \{0.11, \frac{11}{6}, 2\pi, \frac{2e}{3}, -5\frac{1}{2}, -2, 100, 5.\overline{123}, 7, 4.919191\dots, 0.12345\dots\}$$

List the elements of A which belong to the following sets.

- | | | |
|------------------|-------------------|------------------|
| (a) $\mathbb{N}$ | (c) $\mathbb{Q}$ | (e) $\mathbb{R}$ |
| (b) $\mathbb{Z}$ | (d) $\mathbb{Q}'$ | |

3. Write each of the following inequalities using the interval notation and show them on the real number line.

- | | |
|---------------------|---------------------|
| (a) $2 < x < 6$ | (d) $-4 \leq x < 0$ |
| (b) $-3 \leq x < 2$ | (e) $-3 < x$ |
| (c) $-2 < x \leq 0$ | (f) $x \leq 7$ |

4. Show each of the following intervals on the real number line.

- | | |
|--------------------|---------------------|
| (a) $[-2, 3]$ | (d) $(4, \infty)$ |
| (b) $[-1, 5)$ | (e) $(-1, 1)$ |
| (c) $(-\infty, 9]$ | (f) $(-\infty, -3)$ |

5. Write each of the following intervals as inequalities. Use x as the variable

- | | |
|---------------|--------------------|
| (a) $[-5, 2]$ | (d) $[6, \infty)$ |
| (b) $(0, 7)$ | (e) $(-\infty, 2)$ |
| (c) $(-4, 3]$ | (f) $[-7, -2]$ |

6. Write each of the following as single interval.

- (a) $[-5, 2] \cap (-1, 5]$
 (b) $[1, 6) \cap (-3, 4)$
 (c) $\{x : x < 4\} \cup \{x : -1 < x < 6\}$
 (d) $\{x : 2 \leq x \leq 4\} \cup \{x : -1 < x < 5\}$
 (e) $\{x : x > 3\} \cup \{x : -1 < x < 8\}$

7. Find the value of each of the following.

- | | |
|-----------------------------------|--------------------------|
| (a) $(32)^{\frac{1}{5}}$ | (d) $125^{-\frac{4}{3}}$ |
| (b) $(27)^{\frac{2}{3}}$ | (e) $(0.2)^{-3}$ |
| (c) $\frac{16^{\frac{1}{2}}}{49}$ | (f) $(-0.5)^3$ |

8. Simplify each of the following.

- (a) $3^{n+2} \times 9^n \div 27^n$
 (b) $9^{-\frac{1}{2}n} \times 3^{n+3} \times 32^{-\frac{1}{5}}$
 (c) $\frac{\sqrt[3]{a^2b^4}}{\sqrt[3]{a^4b^2}}$
 (d) $\frac{r^{\frac{5}{2}} + r^{\frac{7}{2}}}{r^{\frac{1}{2}}}$
 (e) $\frac{(1+x)^{\frac{1}{2}}}{(1+x)^{\frac{1}{2}} - (1+x)^{-\frac{1}{2}}}$
 (f) $\frac{p^{-1} + q^{-1}}{p^{-2} - q^{-2}}$

9. Solve the following equations.

- (a) $2^{x+1} = \frac{1}{64}$
- (b) $(4^x)^x = 4(8^x)$
- (c) $3^{x+1} - 2(3^x) - 1 = 0$
- (d) $125^x - 25^{x^2+3x-1} = 0$
- (e) $2^{2x+1} - 2^x = 4(2^{x+2}) - 8$
10. (a) If $a^{\frac{1}{2}} - a^{-\frac{1}{2}} = 1$, show that $a + a^{-1} = 3$.
- (b) Given that $z^{\frac{1}{2}} = x^{\frac{1}{2}} + y^{\frac{1}{2}}$. Show that $(x + y - z)^2 = 4xy$.
- (c) If $2^p = 3^q = 12^r$, show that $pq = r(p + 2q)$.
11. Simplify the following.
- (a) $\frac{1}{\sqrt{2}-1} + \frac{1}{\sqrt{2}+1}$
- (b) $\frac{4-3\sqrt{7}}{4+\sqrt{7}} + \frac{4+3\sqrt{7}}{4-\sqrt{7}}$
- (c) $\frac{3\sqrt{5}-4\sqrt{2}}{2\sqrt{5}-3\sqrt{2}}$
- (d) $\frac{1}{\sqrt{3}-\sqrt{2}} + \frac{3}{\sqrt{2}+1} - \frac{2\sqrt{2}}{\sqrt{3}-1}$
12. (a) Find the square roots of each of the following.
- i. $29 + 12\sqrt{5}$
- ii. $11 - 6\sqrt{2}$
- (b) Express $\frac{3+\sqrt{8}}{3-\sqrt{8}}$ in the form $a + \sqrt{b}$, where a and b are integers.
Find the exact value of
- $$\left(\frac{3+\sqrt{8}}{3-\sqrt{8}}\right)^2 + \left(\frac{3-\sqrt{8}}{3+\sqrt{8}}\right)^2.$$
- (c) Solve the following equations.

i. $x + 5 = \sqrt{7-x}$

- ii. $\sqrt{x+6} - \sqrt{x+3} = \sqrt{2x+5}$
- iii. $1 + \sqrt{x+3} = \sqrt{2x-1}$
- iv. $\sqrt{4x-9} + 1 = 2\sqrt{x}$
- v. $\sqrt{x-3} + \sqrt{2x+15} = \sqrt{2-x}$
- vi. $\sqrt{x+4} + \sqrt{x+10} = \sqrt{8x+18}$

13. Solve the following equation, giving the answer correct to 4 significant figures.

- (a) $5(3^x) = 4$
- (b) $2^{3x-4} = 3^{x+1}$
- (c) $6^{3-2x}5^{x+4} = 72$
- (d) $2(2^{2x}) - 2^x - 3 = 0$

14. Solve the following logarithmic equations.

- (a) $2 + \log_3(x-2) = \log_3(x+6)$
- (b) $2 \ln 2x = 4 - \ln 25$
- (c) $3 \log_3 x - 3 \log_x 3 = -8$
- (d) $\log_8(9-x) + \log_8(x+3) = 1\frac{1}{3}$
- (e) $\log_3 x - 4 \log_x 3 + 3 = 0$
- (f) $2 \log 15 + \log(5-x) - \log(4x) = 2$

15. (a) Solve the following simultaneous equations.

$$\begin{aligned}\log_x y + \log_y x &= 2.5 \\ xy &= 27\end{aligned}$$

(b) Given that $2^x = 3^y$ and $x + y = 1$, show that

$$x = \frac{\log 3}{\log 6} \quad y = \frac{\log 2}{\log 6}$$